MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, PROGRAMMABLE VOLTAGE REFERENCES, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, programmable voltage references. Three product assurance classes and a choice of lead material and finishes are provided for each type and are reflected in the part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Programmable voltage reference, 10.000 V, 7.500 V, 5.000 V, 2.500 V (.3%)
02	Programmable voltage reference, 10.000 V, 7.500 V, 5.000 V, 2.500 V (.1%)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline, (see MIL-M-38510, appendix C)
G	A-1 (8-lead can)

1.3 Absolute maximum ratings.

Supply voltage (V_{IN})	40 V
Power dissipation	600 mW
Lead temperature (soldering, 10 seconds)	300°C
Storage temperature	-65° to +175°C
Operating junction temperature (T_J)	-55° to +150°C

1.4 Recommended operating conditions.

Supply voltage range	4.5 V minimum to 30 V maximum
Output current (max)	5 mA
Ambient operating temperature range	-55° to +125°C

1.5 Power and thermal characteristics.

Package	Case outline	Maximum allowable power dissipation 1/	Maximum θ_{J-C} 2/	Maximum θ_{J-A}
8 lead can	G	330 mW @ T_A = 125°C	40°C/W	150°C/W

Notes:
1/ All leads welded or soldered to PC board.
2/ Applies only when $T_C \geq 75°C$.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, RADC (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Terminal connections and block diagram. The terminal connections and block diagram shall be as shown on figure 1.

3.2.2 Schematic circuit. The schematic circuit shall be as specified on figure 2.

3.2.3 Case outline. The case outline shall be as specified in 1.2.3.

3.2.4 Package and sealing material. Package and sealing material shall be in accordance with MIL-M-38510.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

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TABLE I. Electrical performance characteristics.

Characteristic	Symbol	Conditions (unless otherwise specified) $-55°C \le T_A \ge 125°C$, V_{IN} = 15 V, I_L = 0 mA	Device type	Limits Min	Limits Max	Unit
Quiescent current	I_{CC}	V_{IN} = 38 V; V_O = 10 V 1/ T_A = 25°C	01,02	0	1.0	mA
Output voltage	V_{OUT1}	V_O = 10 V T_A = 25°C	01	9.97	10.03	V
		V_O = 10 V	02	9.99	10.01	V
	V_{OUT2}	V_O = 7.5 V	01	7.478	7.522	V
		V_O = 7.5 V	02	7.492	7.508	V
	V_{OUT3}	V_O = 5.0 V	01	4.985	5.015	V
		V_O = 5.0 V	02	4.994	5.006	V
	V_{OUT4}	V_O = 2.5 V	01	2.4925	2.5075	V
		V_O = 2.5 V	02	2.4965	2.5035	V
Line regulation	VR_{LINE1}	12.5 V $\le V_{IN} <$ 15 V; V_O = 10 V	01,02	-.005	.005	%/V
	VR_{LINE2}	15 V $\le V_{IN} \le$ 30 V; V_O = 10 V	01,02	-.002	.002	%/V
	VR_{LINE1}	12.5 V $\le V_{IN} <$ 15 V; V_O = 10 V	01,02	-.010	.010	%/V
	VR_{LINE2}	15 V $\le V_{IN} \le$ 30 V; V_O = 10 V	01,02	-.005	.005	%/V
Load regulation	VR_{LOAD1}	-5 mA $\le I_L \le$ 0 mA; V_O = 10 V	01,02	-50	50	PPM/mA
	VR_{LOAD2}	V_O = 7.5 V				
	VR_{LOAD3}	V_O = 5.0 V				
	VR_{LOAD4}	V_O = 2.5 V				
	VR_{LOAD1}	-5 mA $\le I_L \le$ 0 mA; V_O = 10 V	01,02	-100	100	PPM/mA
	VR_{LOAD2}	V_O = 7.5 V				
	VR_{LOAD3}	V_O = 5.0 V				
	VR_{LOAD4}	V_O = 2.5 V				
Output short circuit current	I_{OS}	V_O = 10 V	01,02	-55	---	mA
Output voltage temperature coefficient		$-55°C \le T_A \le 25°C$ $25°C \le T_A \le 125°C$				
	DV_{OUT1}/DT	V_O = 10 V	01	-0.3	0.3	%FS
	DV_{OUT2}/DT	V_O = 7.5 V				
	DV_{OUT3}/DT	V_O = 5.0 V				
	DV_{OUT4}/DT	V_O = 2.5 V				
	DV_{OUT1}/DT	V_O = 10 V	02	-0.15	0.15	%FS
	DV_{OUT2}/DT	V_O = 7.5 V				
	DV_{OUT3}/DT	V_O = 5.0 V				
	DV_{OUT4}/DT	V_{OUT} = 2.5 V	02	-0.2	0.2	%FS

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions (unless otherwise specified) $-55°C \leq T_A \geq 125°C$, V_{IN} = 15 V, I_L = 0 mA	Device type	Limits		Unit
				Min	Max	
Output noise	N_O	V_O = 10 V; T_A = 25°C; Figures 6 & 7				
		0.1 Hz $\leq$ BW $\leq$ 10 Hz	01,02	---	50	µVp-p
		10 Hz $\leq$ BW $\leq$ 100 kHz	01,02	---	150	µVrms
Settling time to 0.1% of final value (Power up)	t_s(p) (power)	V_O = 10 V; T_A = 25°C; Figure 8				
		I_L = 0 mA	01,02	---	500	µs
		I_L = -5 mA	01,02	---	500	µs

1/ See figure 1 for programming instructions required to obtain the specified values of "V_O".

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3.7 Microcircuit group assignment. The devices covered by this specification shall be in a microcircuit group number 50 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirement	Subgroups (see table III)		
	Class S devices	Class B devices	Class C devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1	None
Final electrical test parameters (method 5004)	1*,2,3	1*,2,3	1
Group A test requirements (method 5005)	1,2,3,4,9	1,2,3,4	1,2,3
Group C end-point and group B, class S electrical parameters (method 5005)	1,2,3, and table IV delta limits	1 and table IV delta limits	1 and table IV delta limits
Additional electrical subgroups for group C periodic inspections	N/A	9	4,9
Group D end-point electrical parameters (method 5005)	1,2,3	1	1

*PDA applies to subgroups 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in tests (method 1015 of MIL-STD-883).

(1) Class S devices: Dynamic tests (test condition D or E) using the circuit shown on figure 4. Test duration shall be 240 hours minimum. Static tests (test condition A) using the circuit shown on figure 4. Test duration shall be 48 hours minimum.

(2) Class B devices: Test condition C or F, using the circuits shown on figure 3 and 5, respectively.

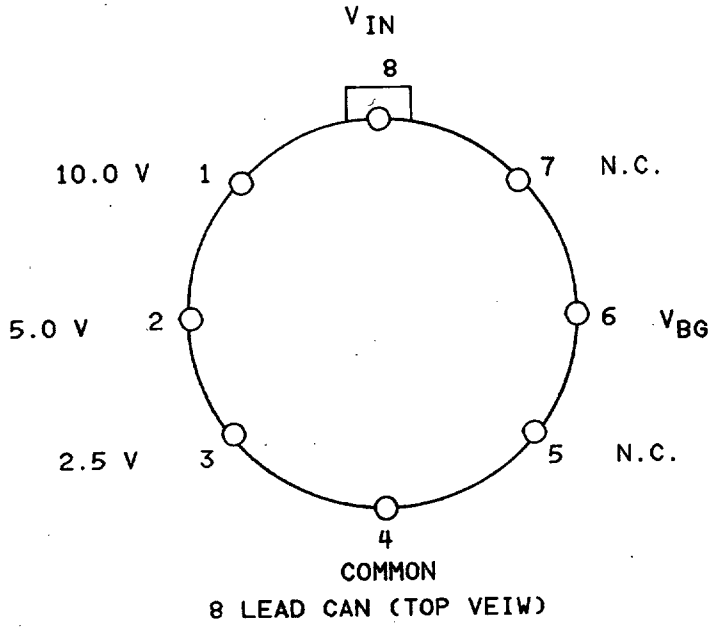
NOTE: If accelerated high-temperture test conditions are used, manufacturer shall ensure that at least 85 percent of the applied voltage is dropped across the device at temperature. The device is not considered functional under accelerated test conditions.

b. Interim and final electrical test parameters shall be as specified in table II herein.

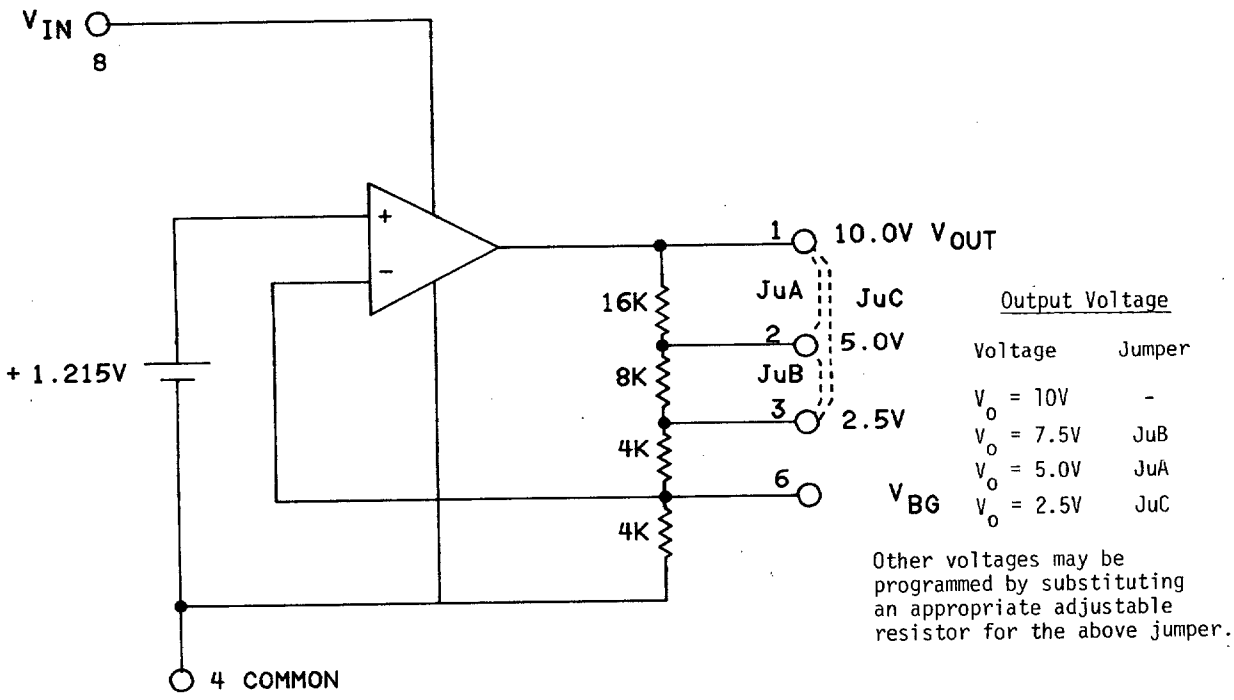
c. The percent defective allowable (PDA) for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as the final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

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Device Types 01, 02 (case G)



Terminal Connections

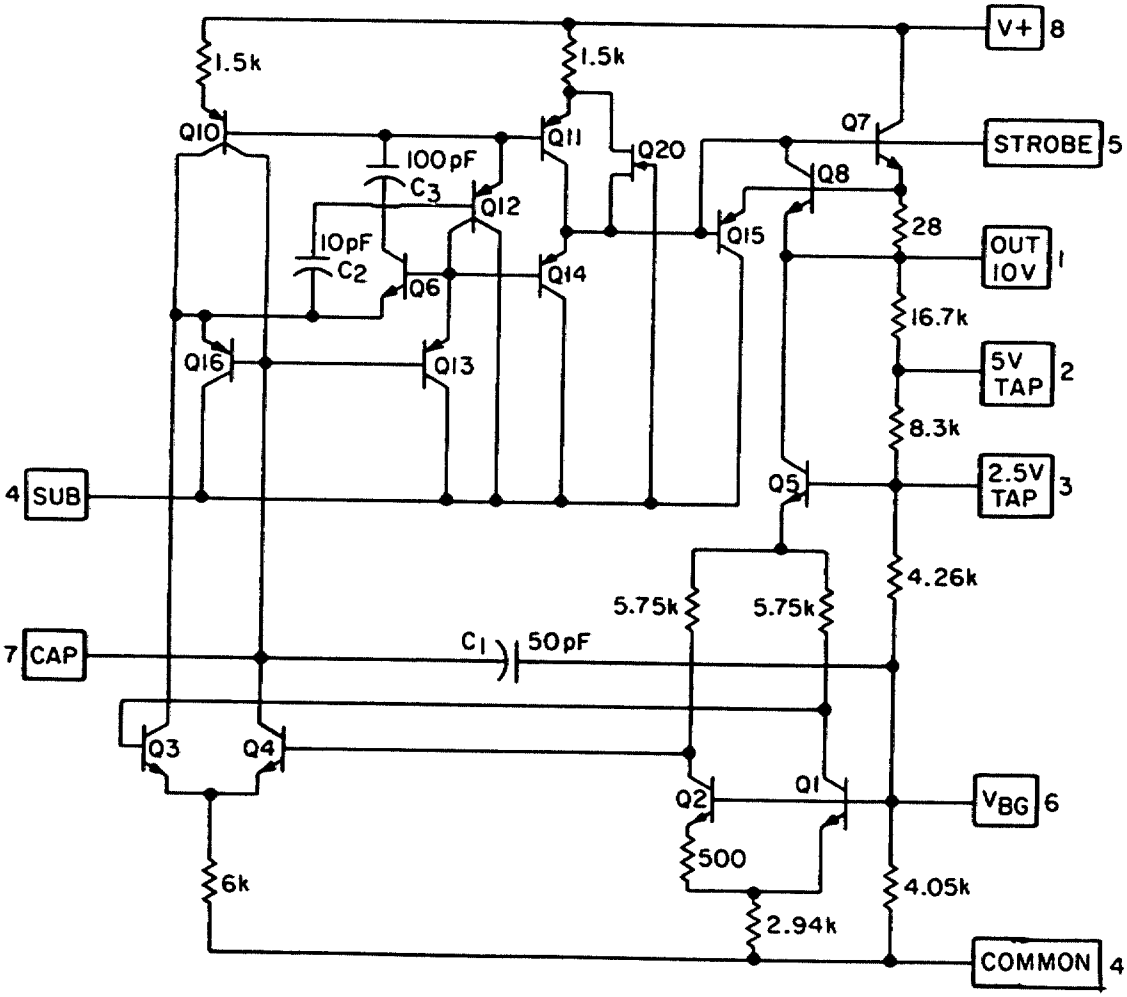


Block diagram.

FIGURE 1. Terminal connections and block diagram.

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Devices 01 and 02

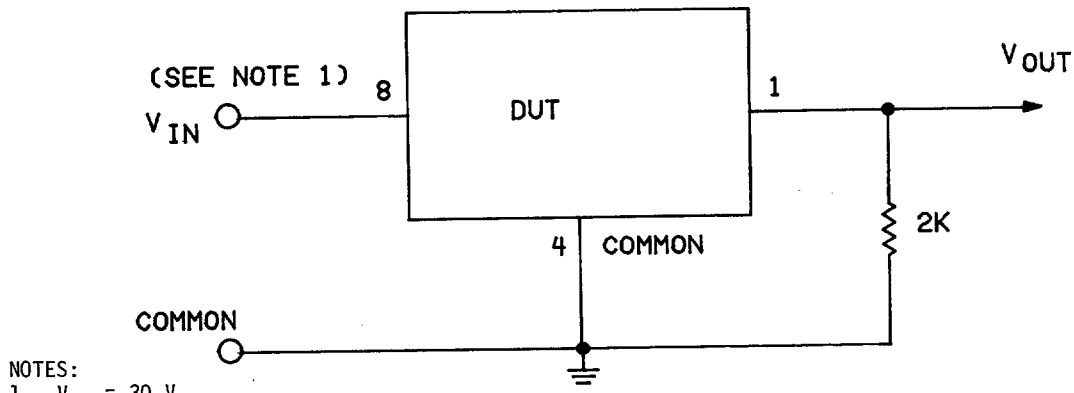


Analog Devices

NOTE: Resistance and capacitance values are nominal. Resistance values are in ohms.

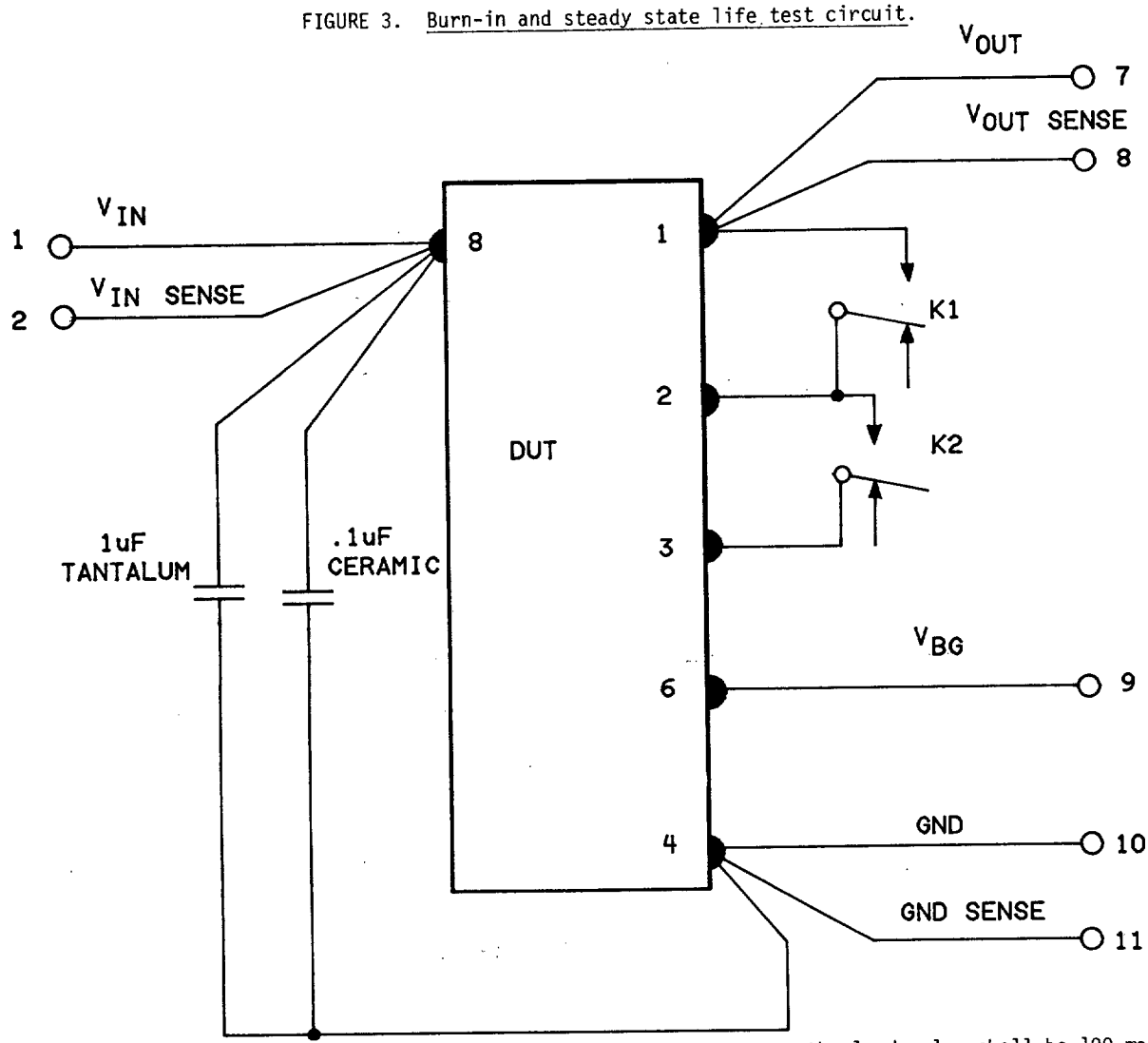
FIGURE 2. Schematic circuits.

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NOTES:
1. V_{IN} = 30 V
2. Test is conducted without a heat sink.

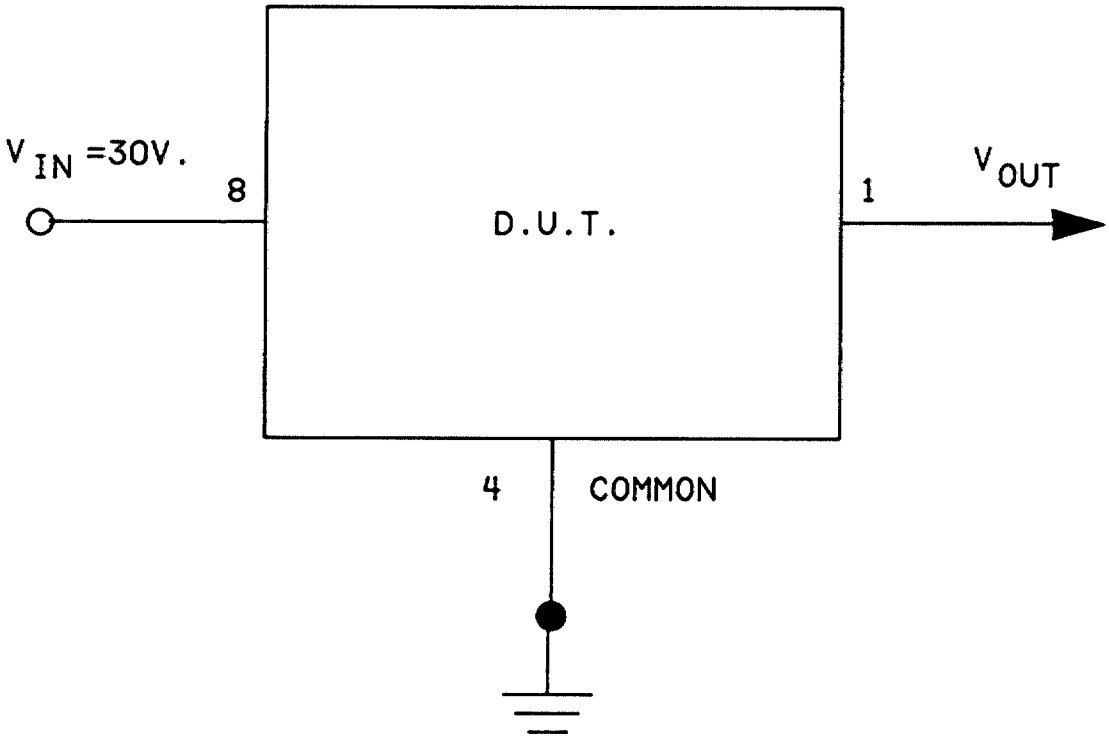
FIGURE 3. Burn-in and steady state life test circuit.



NOTE: 1. For the load regulation test, the load pulse shall be 100 ms maximum, and the duty cycle shall be 1% maximum.

FIGURE 4. Test circuit for static tests.

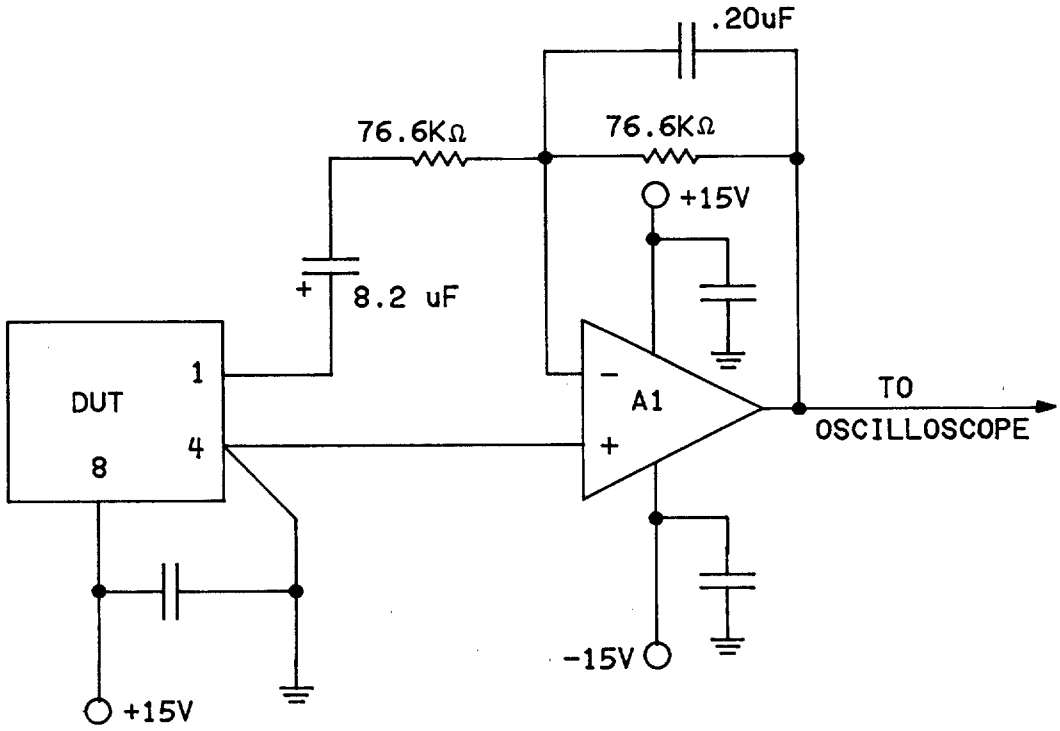
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NOTE: 1. Test is conducted without a heat sink.

FIGURE 5. Accelerated burn-in and steady state life test circuit.

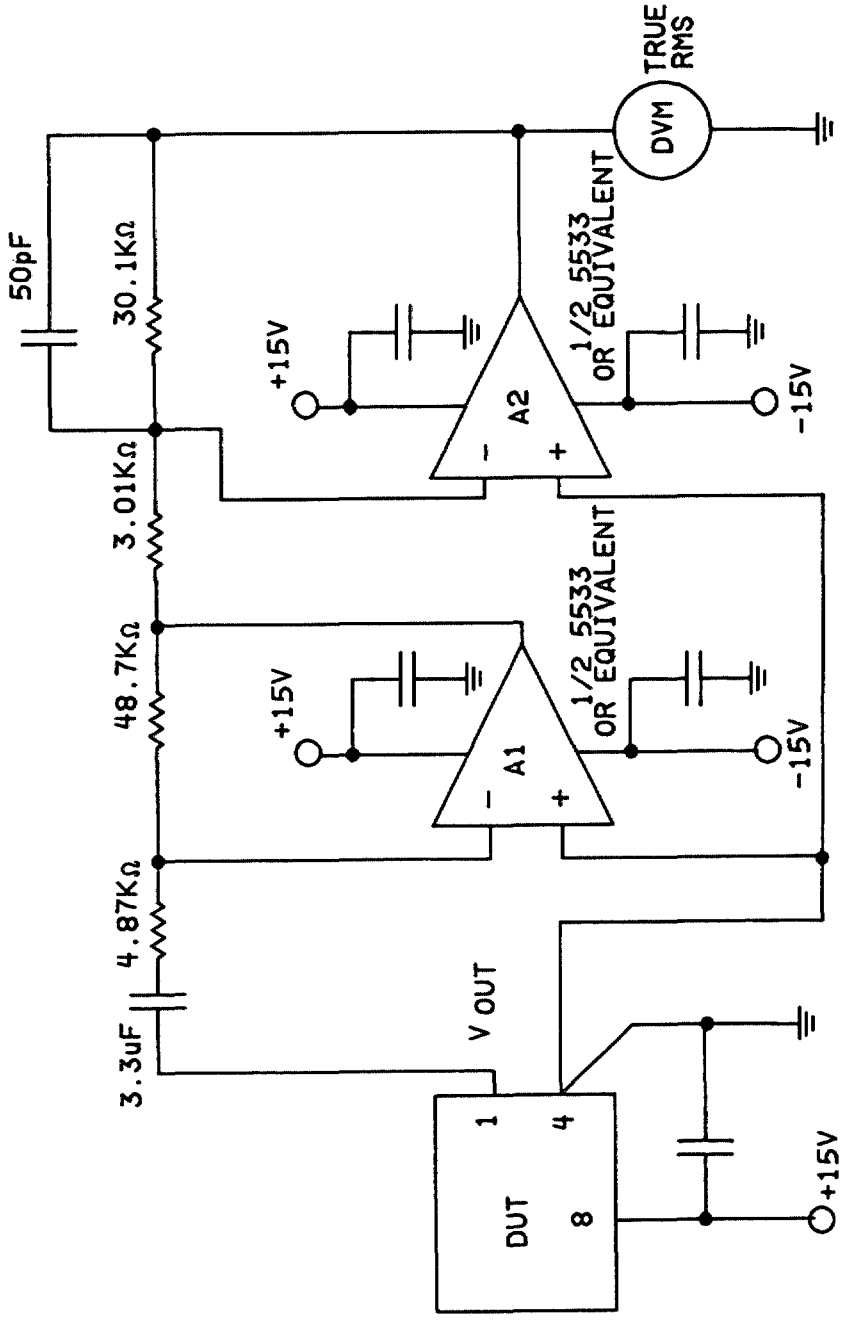
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NOTES:
1. Unless otherwise stated, all capactiors are 47 uF tantalum paralleled with 0.1 uF ceramic.
2. The oscilloscope pre-amp shall be a d-c voltage comparator type.
3. The oscilloscope time base shall be set to 1.0 Hz/cm and the horizontal display shall be 10 cm long.

FIGURE 6. Otuput "1/f" noise test circuit.

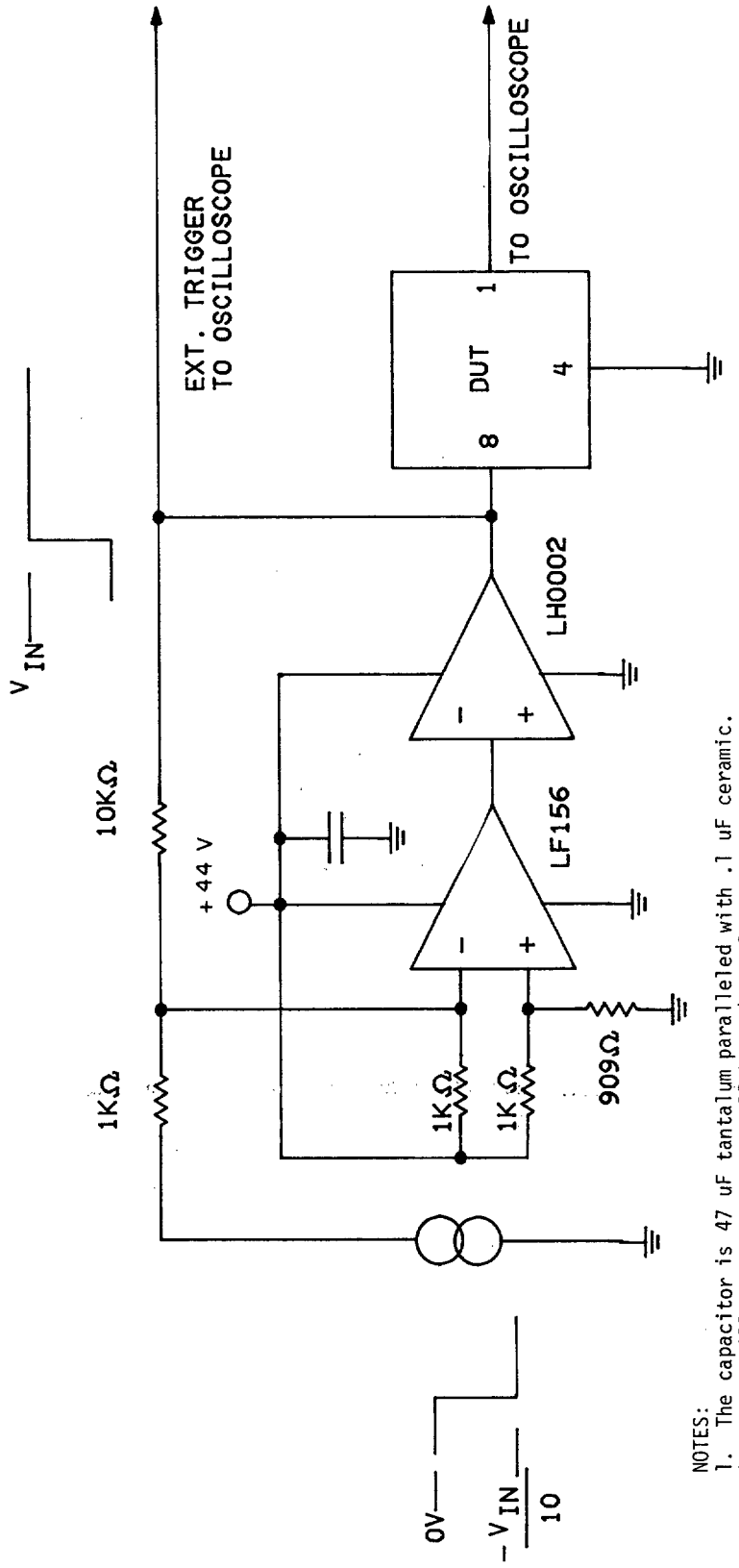
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NOTES:
1. Unless otherwise stated, all capacitors are 47 uF tantalum paralleled with 0.1 uF ceramic.
2. The meters shall be true rms reading and shall have a bandwidth greater than 1 Meg Hz.
3. A1 and A2 shall have a gain bandwidth product greater than 10 MHz and an output noise density less than 10 nV/Hz.

FIGURE 7. Output "white" noise test circuit.

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NOTES:
1. The capacitor is 47 uF tantalum paralleled with .1 uF ceramic.
2. The oscilloscope pre-amp shall be a d-c voltage comparator type.
3. The settling time is measured at the point in time where the device output voltage is within 0.1% of its final value.
4. The oscilloscope bandwidth shall be greater than 2 MHz.

FIGURE 8. Power up settling time test circuit.

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TABLE III. Group A inspection for device types 01 and 02.

Sub group	Symbol	Test No.	Test conditions (figure 4)		Applied voltage and current reference pin 10			Relays energized	Measurement sense lines reference pin 11			Equations	Device type	Limits		Unit
			V_{IN} (V)	I_L (mA)	V_O (V)	1 (V)	7 (mA)		Pin	Value	Unit			Min	Max	
1 T_A = 25°C	I_{CC1}	1	40	0	10	40	---	---	1	I1	mA	I_{CC1} = I1	01,02	0	1.0	mA
	I_{CC2}	2	15	↓	10	15	---	---	1	I2	mA	I_{CC2} = I2	01,02	0	1.0	mA
	V_{OUT1}	3	15		10	↓	---	---	8	E1	V	V_{OUT1} = E1	01	9.97	10.03	V
													02	9.99	10.01	
	V_{OUT2}	4	↓		7.5		---	K2	↓	E2	↓	V_{OUT2} = E2	01	7.478	7.522	↓
													02	7.492	7.508	
	V_{OUT3}	5	↓		5.0		---	K1	↓	E3	↓	V_{OUT3} = E3	01	4.985	5.015	↓
													02	4.994	5.006	
	V_{OUT4}	6	↓		2.5	↓	---	K1,K2	↓	E4	↓	V_{OUT4} = E4	01	2.4925	2.5075	↓
													02	2.4965	2.5035	
	VR_{LINE1}	7	12.5	↓	10	12.5	---	---	↓	E5	↓	VR_{LINE1} = 100 (E1-E5)/(2.5E1)	01,02	-.005	.005	%/V
	VR_{LINE2}	8	30		10	30	---	---		E6		VR_{LINE2} = 100 (E6-E1)/(15E1)	01,02	-.002	.002	%/V
	VR_{LOAD1}	9	15	-5	10	15	-5	---	↓	E7	↓	VR_{LOAD1} = (E1-E7) (10^6)/(5E1)	01,02	-50	50	PPM/mA
	VR_{LOAD2}	10	↓	↓	7.5	↓	↓	K2		E8		VR_{LOAD2} = (E2-E8) (10^6)/(5E2)	01,02	↓	↓	↓
	VR_{LOAD3}	11			5.0			K1		E9		VR_{LOAD3} = (E3-E9) (10^6)/(5E3)	01,02			
	VR_{LOAD4}	12			2.5			K1,K2		E10		VR_{LOAD4} = (E4-E10) (10^6)/(5E4)	01,02			
	I_{OS1}	13	↓	---	10	↓	---	---	7	I3	mA	I_{OS1} = I3	01,02	-55	---	mA
2 T_A = 125°C	DV_{OUT1}/DT	15		0	10		---	---	8	E11	V	DV_{OUT1}/DT = (E1-E11) (10^2)/(E1)	01	-0.3	0.3	%FS
													02	-0.15	0.15	
	DV_{OUT2}/DT	16	↓	↓	7.5	↓	---	K2	↓	E12	↓	DV_{OUT2}/DT = (E2-E12) (10^2)/(E2)	01	-0.3	0.3	↓
													02	-0.15	0.15	
	DV_{OUT3}/DT	17	↓		5.0	↓	---	K1	↓	E13	↓	DV_{OUT3}/DT = (E3-E13) (10^2)/(E3)	01	-0.3	0.3	↓
													02	-0.15	0.15	
	DV_{OUT4}/DT	18	↓		2.5	↓	---	K1,K2	↓	E14	↓	DV_{OUT4}/DT = (E4-E14) (10^2)/(E4)	01	-0.3	0.3	↓
													02	-0.2	0.2	

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Sub group	Symbol	Test No.	Test conditions (figure 4) Applied voltage and current reference pin 10					Relays energized	Measurement sense lines reference pin 11			Equations	Device type	Limits		Unit
			V_{IN} (V)	I_L (mA)	V_o (V)	1 (V)	7 (mA)		Pin	Value	Unit			Min	Max	
2 T_A= 125°C	VR_{LINE1} VR_{LINE2}	19 20	12.5 30	0 0	10 10	12.5 30	--- ---	--- ---	8	E15 E16	V	VR_{LINE1} = 100 (E11-E15)/(2.5E11) VR_{LINE2} = 100 (E16-E11)/(15E11)	01,02 01,02	-0.010 -.005	0.010 .005	%/V %/V
	VR_{LOAD1}	21	15	-5	10	15	-5	---		E17		VR_{LOAD1} = (E11-E17) (10^6)/(5E11)	01,02	-100	100	PPM/mA
	VR_{LOAD2}	22			7.5			K2		E18		VR_{LOAD2} = (E12-E18) (10^6)/(5E12)	01,02			
	VR_{LOAD3}	23			5			K1		E19		VR_{LOAD3} = (E13-E19) (10^6)/(5E13)	01,02			
	VR_{LOAD4}	24			2.5			K1,K2		E20		VR_{LOAD4} = (E14-E20) (10^6)/(5E14)	01,02			
	I_{OS1}	25		---	10		---	---	7	I4	mA	I_{OS1} = I4	01,02	-55	---	mA
3 T_A= -55°C	DV_{OUT1}/DT	26		0	10		---	---	8	E21	V	DV_{OUT1}/DT = (E1-E21) (10^2)/(E1)	01 02	-0.3 -0.15	0.3 0.15	%FS
	DV_{OUT2}/DT	27			7.5		---	K2		E22		DV_{OUT2}/DT = (E2-E22) (10^2)/(E2)	01 02	-0.3 -0.15	0.3 0.15	
	DV_{OUT3}/DT	28			5		---	K1		E23		DV_{OUT3}/DT = (E3-E23) (10^2)/(E3)	01 02	-0.3 -0.15	0.3 0.15	
	DV_{OUT4}/DT	29			2.5		---	K1,K2		E24		DV_{OUT4}/DT = (E4=E24) (10^2)/(E4)	01 02	-0.3 -0.2	0.3 0.2	
	VR_{LINE1} VR_{LINE2}	30 31	12.5 30		10	12.5 30	--- ---	--- ---		E25 E26		VR_{LINE1} = 100 (E21-E25)/(2.5E21) VR_{LINE2} = 100 (E26-E21)/(15E21)	01,02 01,02	-0.010 -.005	0.010 .005	%/V %/V
	VR_{LOAD1} VR_{LOAD2} VR_{LOAD3} VR_{LOAD4}	32 33 34 35	15	-5	10 7.5 5 2.5	15	-5	--- K2 K1 K1, K2		E27 E28 E29 E30		VR_{LOAD1} = (E21-E27) (10^6)/(5E21) VR_{LOAD2} = (E22-E28) (10^6)/(5E22) VR_{LOAD3} = (E23-E29) (10^6)/(5E23) VR_{LOAD4} = (E24-E30) (10^6)/(5E24)	01,02 01,02 01,02 01,02	-100	100	PPM/mA
	I_{OS1}	36		---	10		---	---	7	I5	mA	I_{OS1} = I5	01,02	-55	---	mA

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Sub group	Symbol	Test No.	Test conditions (figure 4) Applied voltage and current reference pin 10					Relays energized	Measurement sense lines reference pin 11			Equations	Device type	Limits		Unit
			V_{IN} (V)	I_L (mA)	V_O (V)	I (V)	I (mA)		Pin	Value	Unit			Min	Max	
4 T_A = 25°C	N_O	37	15	0	10	15	---	---	---	E31	µVp-p	N_O = E31 (see figure 6)	01,02	---	50	µVp-p
		38	15	0	10	15	---	---	---	E32	µVrms	N_O = E32 (see figure 7)	01,02	---	150	µVrms
9 T_A = 25°C	$t_s(p)$	39	15	0	10	15	---	---	1	t1	µs	$t_s(p)$ = t1 (see figure 8)	01,02	---	500	µs

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4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 5, 6, 7, 8, 10, and 11 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Electrical parameters shall be as specified in table II herein.

b. Life test for class S devices shall be in accordance with table IIa (subgroup 5) of method 5005 of MIL-STD-883, using the circuit shown on figure 3. If the alternate burn-in conditions are used, the circuits on figure 5 shall be used.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection for class B and C devices.

b. Steady-state life tests for class B and C devices (method 1005 of MIL-STD-883) - Test condition C, using the circuits shown on figure 3, or test condition F, using the circuits shown on figure 5.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

TABLE IV. Group C end-point electrical parameters.

Device type	Characteristic	Symbol	Delta Limits 1/	Limits Min	Limits Max	Unit
01	Output voltage 1 V_O = 10 V	V_{OUT1}	±0.05%	9.97	10.03	V
02	V_O = 10 V			9.99	10.01	
01	Output voltage 2 V_O = 7.5 V	V_{OUT2}		7.478	7.522	
02	V_O = 7.5 V			7.492	7.508	
01	Output voltage 3 V_O = 5.0 V	V_{OUT3}		4.985	5.015	
02	V_O = 5.0 V			4.994	5.006	
01	Output voltage 4 V_O = 2.5 V	V_{OUT4}		2.4925	2.5075	
02	V_O = 2.5 V			2.4965	2.5035	

1/ Delta limits apply to the measured value (see delta limit definition in MIL-M-38510).

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate table and as follows:

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4.5.1 Voltage and current. All voltages given are referenced to the pin 3 ground terminal of the device under test (DUT). Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cooldown procedure. When these tests are completed and prior to removal of bias voltages, the device under test shall be cooled to an ambient temperature of 25°C ±3°C, and the electrical parameter end-point measurements shall then be performed.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product to the contracting activity in addition to notification to qualifying activity, if applicable.

e. Requirements for packaging and packing.

f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

g. Requirements for product assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

i. Requirements for JAN marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

6.4.1 Line regulation (VR_{LINE}). Line regulation is the change in the output voltage for a specified change in the input voltage.

6.4.2 Load regulation (VR_{LOAD}). Load regulation is the change in the output voltage for a specified change in the load current.

6.4.3 Output noise (N_O). Output noise is the electrical noise, measured at the output, with a constant load current and a constant input voltage. The "1/f" noise is the peak-to-peak noise voltage measured over a frequency band from 0.1 Hz to 10 Hz. The "white" noise is the rms noise voltage measured over a frequency band from 10 Hz to 100 kHz.

6.4.4 Settling time (t_s). Settling time is the time for the output voltage to settle to within 0.1% of its final value.

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6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	584S
02	584T

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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